

Estimating Bayesian IRT Models in Stan

SMIP Summer School 2025: Lecture 05

Section Objectives

1. Show how to estimate unidimensional latent variable models with dichotomous data
 - Also known as Item Response Theory (IRT) or Item Factor Analysis (IFA) models
2. Show how to estimate different parameterizations of IRT/IFA models
3. Describe how to obtain IRT/IFA auxiliary statistics from Markov Chains
4. Show variations of various dichotomous-data models

Dichotomous Data Distribution: Bernoulli

The Bernoulli Distribution

The Bernoulli distribution is a one-trial version of the Binomial distribution

- Sample space (support) $Y \in \{0, 1\}$

The probability mass function (pdf):

$$P(Y = y) = \pi^y(1 - \pi)^{(1-y)}$$

The distribution has only one parameter: π (the probability $Y = 1$)

- Mean of the distribution: $E(Y) = \pi$
- Variance of the distribution: $Var(Y) = \pi(1 - \pi)$

Definition: Dichotomous vs. Binary

Note the definitions of some of the words for data with two values:

- Dichotomous: Taking two values (without numbers attached)
- Binary: either zero or one (specifically: $\{0, 1\}$)

Therefore:

- Not all dichotomous variables are binary (i.e., $\{2, 7\}$ is a dichotomous variable)
- All binary variables are dichotomous

Finally:

- Bernoulli distributions are for binary variables
- Most dichotomous variables can be recoded as binary variables without loss of model effects

Models with Bernoulli Distributions

Generalized linear models using Bernoulli distributions put a linear model onto a transformation of the mean

- Link functions map the mean $E(Y)$ from its original range of $[0, 1]$ to $(-\infty, \infty)$

For an unconditional (empty) model, this is shown here:

$$f(E(Y)) = f(\pi)$$

Link Functions for Bernoulli Distributions

Common choices for the link function (in latent variable models):

- Logit (or log odds):

$$f(\pi) = \log\left(\frac{\pi}{1 - \pi}\right)$$

- Probit:

$$f(\pi) = \Phi^{-1}(\pi)$$

Where Φ is the inverse cumulative distribution of a standard normal distribution:

$$\Phi(Z) = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right) dx$$

Less Common Link Functions

In the generalized linear models literature, there are a number of different link functions:

- Log-log: $f(\pi) = -\log(-\log(\pi))$
- Complementary Log-Log: $f(\pi) = \log(-\log(1 - \pi))$

Most of these seldom appear in latent variable models

- Each has a slightly different curve shape

Inverse Link Functions

Our latent variable models will be defined on the scale of the link function

- Sometimes we wish to convert back to the scale of the data
 - Example: Test characteristic curves mapping θ_p onto an expected test score

For this, we need the inverse link function

- Logit (or log odds) link function:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1 - \pi}\right)$$

- Logit (or log odds) inverse link function:

$$\pi = \frac{\exp(\text{logit}(\pi))}{1 + \exp(\text{logit}(\pi))} = \frac{1}{1 + \exp(-\text{logit}(\pi))} = (1 + \exp(-\text{logit}(\pi)))^{-1}$$

Latent Variable Models with Bernoulli Distributions for Observed Variables

Latent Variable Models with Bernoulli Distributions for Observed Variables

We define a latent variable model for binary responses using a Bernoulli distribution

- To start, we will use the logit link function
- We will begin with the linear predictor we had from the normal distribution models (Confirmatory factor analysis: $\mu_i + \lambda_i \theta_p$)

Latent Variable Models with Bernoulli Distributions for Observed Variables

For an item i and a person p , the model becomes:

$$\text{logit}\left(P\left(Y_{pi} = 1\right)\right) = \mu_i + \lambda_i\theta_p$$

- Note: the mean π_i is replaced by $P\left(Y_{pi} = 1 \mid \theta_p\right)$
 - This is the mean of the observed variable, conditional on θ_p
- The item intercept is μ_i : The expected logit when $\theta_p = 0$
- The item discrimination is λ_i : The change in the logit for a one-unit increase in θ_p

Model Family Names

Depending on your field, the model from the previous slide can be called:

- The two-parameter logistic (2PL) model with slope/intercept parameterization
- An item factor model

Model Family Names

These names reflect the terms given to the model in diverging literatures:

- 2PL: Educational measurement
 - Birnbaum, A. (1968). Some Latent Trait Models and Their Use in Inferring an Examinee's Ability. *In F. M. Lord & M. R. Novick (Eds.), Statistical Theories of Mental Test Scores* (pp. 397-424). Reading, MA: Addison-Wesley.
- Item factor analysis: Psychology
 - Christofferson, A.(1975). Factor analysis of dichotomous variables. *Psychometrika*, 40, 5-22.

Estimation methods are the largest difference between the two families

Differences from Normal Distributions

Recall our normal distribution models:

$$Y_{pi} = \mu_i + \lambda_i \theta_p + e_{p,i}, \quad e_{p,i} \sim N(0, \psi_i^2)$$

Compared to our Bernoulli distribution models:

$$\text{logit}\left(P(Y_{pi} = 1)\right) = \mu_i + \lambda_i \theta_p$$

Differences:

- No residual (unique) variance ϕ^2 in Bernoulli distribution
 - Only one parameter in distribution; variance is a function of the mean
- Identity link function in normal distribution: $f(E(Y_{pi} | \theta_p)) = E(Y_{pi} | \theta_p)$
 - Model scale and data scale are the same
- Logit link function in Bernoulli distribution

From Model Scale to Data Scale

Commonly, the IRT or IFA model is shown on the data scale (using the inverse link function):

$$P(Y_{pi} = 1) = \frac{\exp(\mu_i + \lambda_i \theta_p)}{1 + \exp(\mu_i + \lambda_i \theta_p)}$$

The core of the model (the terms in the exponent on the right-hand side) is the same

- Models are equivalent
 - $P(Y_{pi} = 1)$ is on the data scale
 - $\text{logit}(P(Y_{pi} = 1))$ is on the model (link) scale

Modeling All Data

$$\text{logit}(Y_{p1} = 1) = \mu_1 + \lambda_1 \theta_p$$

$$\text{logit}(Y_{p2} = 1) = \mu_2 + \lambda_2 \theta_p$$

$$\text{logit}(Y_{p3} = 1) = \mu_3 + \lambda_3 \theta_p$$

$$\text{logit}(Y_{p4} = 1) = \mu_4 + \lambda_4 \theta_p$$

$$\text{logit}(Y_{p5} = 1) = \mu_5 + \lambda_5 \theta_p$$

$$\text{logit}(Y_{p6} = 1) = \mu_6 + \lambda_6 \theta_p$$

$$\text{logit}(Y_{p7} = 1) = \mu_7 + \lambda_7 \theta_p$$

$$\text{logit}(Y_{p8} = 1) = \mu_8 + \lambda_8 \theta_p$$

$$\text{logit}(Y_{p9} = 1) = \mu_9 + \lambda_9 \theta_p$$

$$\text{logit}(Y_{p10} = 1) = \mu_{10} + \lambda_{10} \theta_p$$

Measurement Model Analysis Steps

1. Specify model
2. Specify scale identification method for latent variables
3. Estimate model
4. Examine model-data fit
5. Iterate between steps 1-4 until adequate fit is achieved

Measurement Model Auxiliary Components

6. Score estimation (and secondary analyses with scores)
7. Item evaluation
8. Scale construction
9. Equating
10. Measurement invariance/differential item functioning

Model Specification

The set of equations on the previous slide formed step #1 of the Measurement Model Analysis Steps:

1. Specify Model

The next step is:

2. Specify scale identification method for latent variables

We will initially assume $\theta_p \sim N(0, 1)$, which allows us to estimate all item parameters of the model

- This is what we call a standardized latent variable
 - They are like Z-scores

Identification of Latent Traits, Part 1

Psychometric models require two types of identification to be valid:

1. Empirical Identification

- The minimum number of items that must measure each latent variable
- From CFA: three observed variables for each latent variable (or two if the latent variable is correlated with another latent variable)

Bayesian priors can help to make models with fewer items than these criteria suggest estimable

- The parameter estimates (item parameters and latent variable estimates) often have MCMC convergence issues and should not be trusted
- Use the CFA standard in your work

Identification of Latent Traits, Part 2

Psychometric models require two types of identification to be valid:

2. Scale Identification (i.e., what the mean/variance is for each latent variable)

- The additional set of constraints needed to set the mean and standard deviation (variance) of the latent variables
- Two main methods to set the scale:
 - Marker item parameters
 - For variances: Set the loading/slope to one for one observed variable per latent variable
 - Can estimate the latent variable's variance (the diagonal of Σ_{θ})
 - For means: Set the item intercept to one for one observed variable per latent variable
 - Can estimate the latent variable's mean (in μ_{θ})

Identification of Latent Traits, Part 2

2. Scale Identification (i.e., what the mean/variance is for each latent variable)
 - Standardized factors
 - Set the variance for all latent variables to one
 - Set the mean for all latent variables to zero
 - Estimate all unique off-diagonal correlations (covariances) in Σ_{θ}

More on Scale Identification

Bayesian priors can let you believe you can estimate more parameters than the non-Bayesian standards suggest

- For instance, all item parameters and the latent variable means/variances

Like empirical identification, these estimates are often unstable and are not recommended

- Standardized latent variables
 - Used for scale development and/or when scores are of interest directly
- Marker item for latent variables and zero means
 - Used for cases where latent variables may become outcomes (and that variance needs explained)

Important Point: Regardless of model choice, model/data likelihoods are equivalent

- Differing prior distributions may make models non-equivalent

Model (Data) Likelihood Functions

The specification of the model defines the model (data) likelihood function for each type of parameter

- To demonstrate, let's examine the data likelihood for the factor loading for the first item λ_1

The model (data) likelihood function can be defined conditional on all other parameter values (as in a block in an MCMC iteration)

- That is: hold μ_1 and θ constant

The likelihood is then:

$$f(Y_{p1} \mid \lambda_1) = \prod_{p=1}^P (\pi_{p1})^{Y_{p1}} (1 - \pi_{p1})^{1 - Y_{p1}}$$

Model (Data) Log Likelihood Functions

As this number can be very small (making numerical precision an issue), we often take the log:

$$\log f(Y_{p1} \mid \lambda_1) = \sum_{p=1}^P \log \left[(\pi_{p1})^{Y_{p1}} (1 - \pi_{p1})^{1 - Y_{p1}} \right]$$

Model (Data) Log Likelihood Functions

The key in the likelihood function is to substitute each person's data-scale model for π_{p1} :

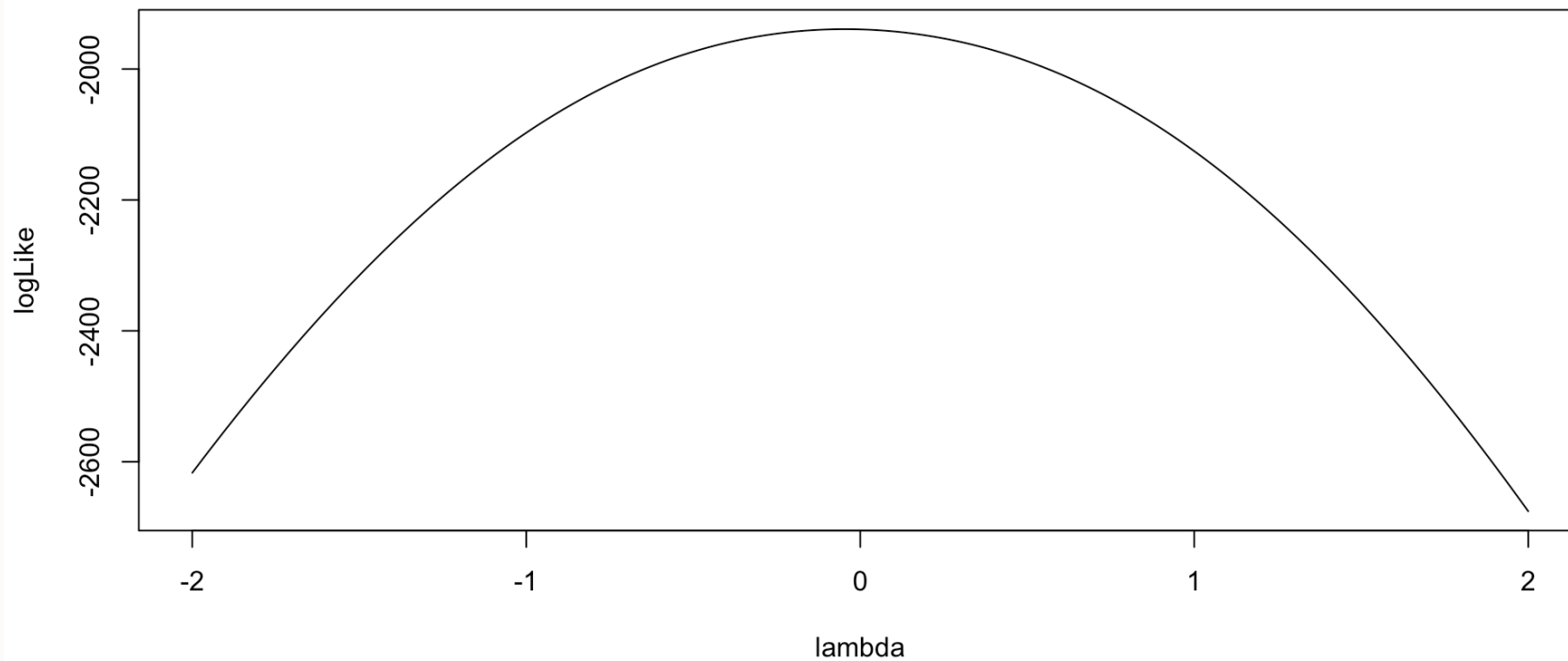
$$\pi_{p1} = \frac{\exp(\mu_1 + \lambda_1 \theta_p)}{1 + \exp(\mu_1 + \lambda_1 \theta_p)}$$

Which then becomes:

$$\log f(Y_{p1} | \lambda_1) = \sum_{p=1}^P \log \left[\left(\frac{\exp(\mu_1 + \lambda_1 \theta_p)}{1 + \exp(\mu_1 + \lambda_1 \theta_p)} \right)^{Y_{p1}} \left(1 - \frac{\exp(\mu_1 + \lambda_1 \theta_p)}{1 + \exp(\mu_1 + \lambda_1 \theta_p)} \right)^{1 - Y_{p1}} \right]$$

Model (Data) Log Likelihood Functions

As an example for λ_1 :



Model (Data) Log Likelihood Functions for θ_p

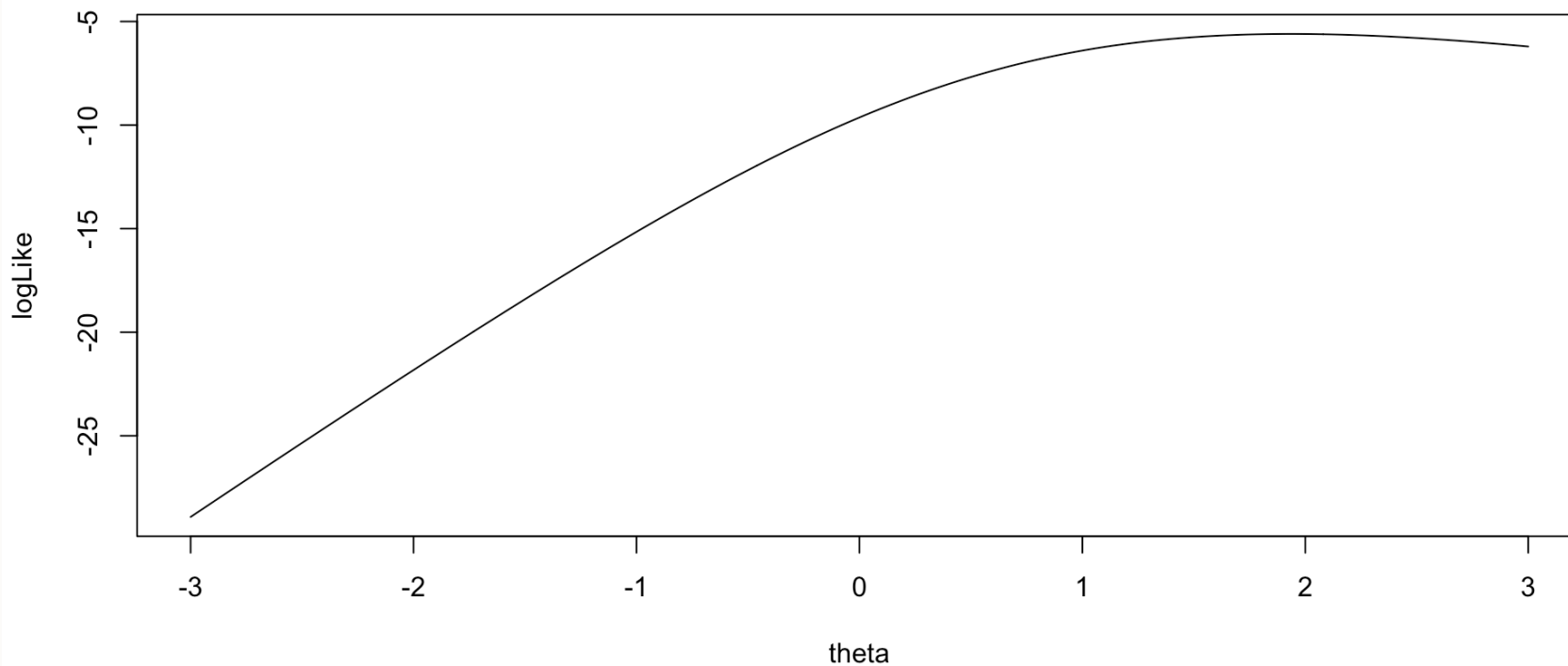
For each person, the same model (data) likelihood function is used

- Only now it varies across each item response
- Example: Person 1

$$f(Y_{1i} | \theta_1) = \prod_{i=1}^I (\pi_{1i})^{Y_{1i}} (1 - \pi_{1i})^{1 - Y_{1i}}$$

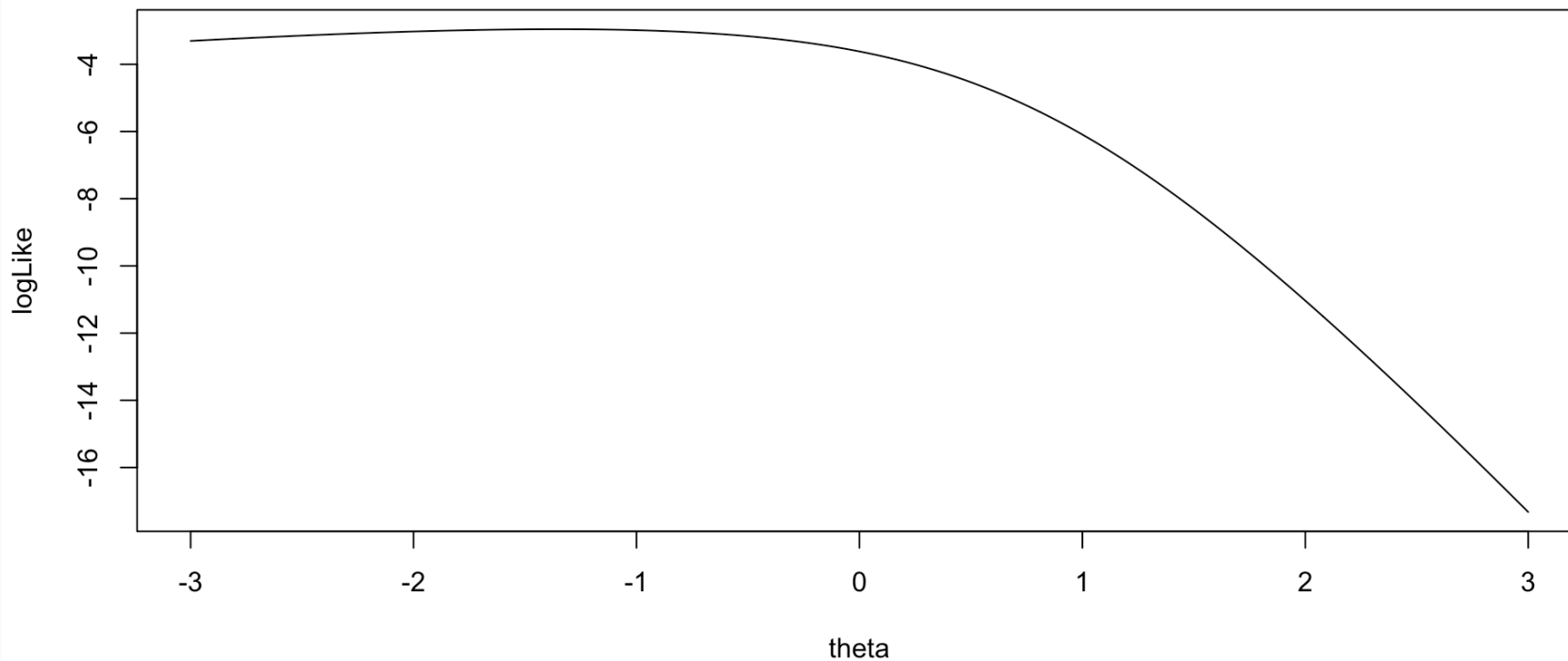
Model (Data) Log Likelihood Functions

As an example for the log-likelihood for θ_2 :



Model (Data) Log Likelihood Functions

As an example for the log-likelihood for θ_1 :



Implementing Bernoulli Outcomes in Stan

Stan's `model` Block

```
1 model {  
2  
3   lambda ~ multi_normal(priorMeanLambda, priorCovLambda); // Prior for item discrimination/factor loadings  
4   mu ~ multi_normal(priorMeanMu, priorCovMu); // Prior for item intercepts  
5  
6   theta ~ normal(0, 1); // Prior for latent variable (with mean/sd specified)  
7  
8   for (item in 1:nItems){  
9     Y[item] ~ bernoulli_logit(mu[item] + lambda[item]*theta);  
10  }  
11  
12 }
```

For logit models without lower/upper asymptote parameters, Stan has a convenient `bernoulli_logit()` function

- Automatically has the link function embedded
- The catch: The data have to be defined as an integer

Also, note that there are few differences from the normal outcomes models (CFA)

- No residual variance parameters

Stan's parameters Block

```
1 parameters {  
2   vector[nObs] theta;           // the latent variables (one for each person)  
3   vector[nItems] mu;           // the item intercepts (one for each item)  
4   vector[nItems] lambda;       // the factor loadings/item discriminations (one for each item)  
5 }
```

Stan's `data {}` Block

```
1 data {  
2   int<lower=0> nObs;           // number of observations  
3   int<lower=0> nItems;        // number of items  
4   array[nItems, nObs] int<lower=0, upper=1> Y; // item responses in an array  
5  
6   vector[nItems] priorMeanMu; // prior mean vector for intercept parameters  
7   matrix[nItems, nItems] priorCovMu; // prior covariance matrix for intercept parameters  
8  
9   vector[nItems] priorMeanLambda; // prior mean vector for loading parameters  
10  matrix[nItems, nItems] priorCovLambda; // prior covariance matrix for loading parameters  
11 }
```

One difference from normal outcomes model—the data are defined as an `array` :

```
1 array[nItems, nObs] int<lower=0, upper=1> Y;
```

- Arrays are types of matrices (with more than two dimensions possible)
 - Allows for different types of data (here Y are integers)
 - Integer-valued variables needed for `bernoulli_logit()` function
- Arrays are row-major (meaning order of items and persons is switched)

Change to Data List for Stan Import

The switch of items and observations in the `array` statement means the data imported have to be transposed:

```
1  # build stan data file
2  model2PL_Data = list(
3    nObs = nrow(modelingData),
4    nItems = 10,
5    Y = t(correctResponseData),
6    priorMeanMu = rep(0, nItems),
7    priorCovMu = 10 * diag(nItems),
8    priorMeanLambda = rep(0, nItems),
9    priorCovLambda = 10 * diag(nItems)
10 )
```

Running the Model In Stan

The Stan program takes longer to run than in linear models:

- Note: Typically, longer chains are needed for larger models like this
- Note: Starting values added (mean of 5 is due to logit function limits)
 - Helps keep definition of parameters (stay away from opposite mode)
 - Too large of value can lead to NaN values (exceeding numerical precision)

See `lecture05_unidimensionalBayesianIRT.R` for syntax and results

Modeling vs. Didactic Strategy

At this point, one should investigate model fit of the model we just ran

- If the model does not fit, then all model parameters could be biased
 - Both item parameters and person parameters (θ_p)
 - Moreover, the uncertainty accompanying each parameter (the posterior standard deviation) may also be biased
 - Especially bad for psychometric models as we quantify reliability with these numbers
-

But, to teach generalized measurement models, we will first talk about differing models for observed data

- Different distributions
- Different parameterizations across the different distributions

Investigating Item Parameters

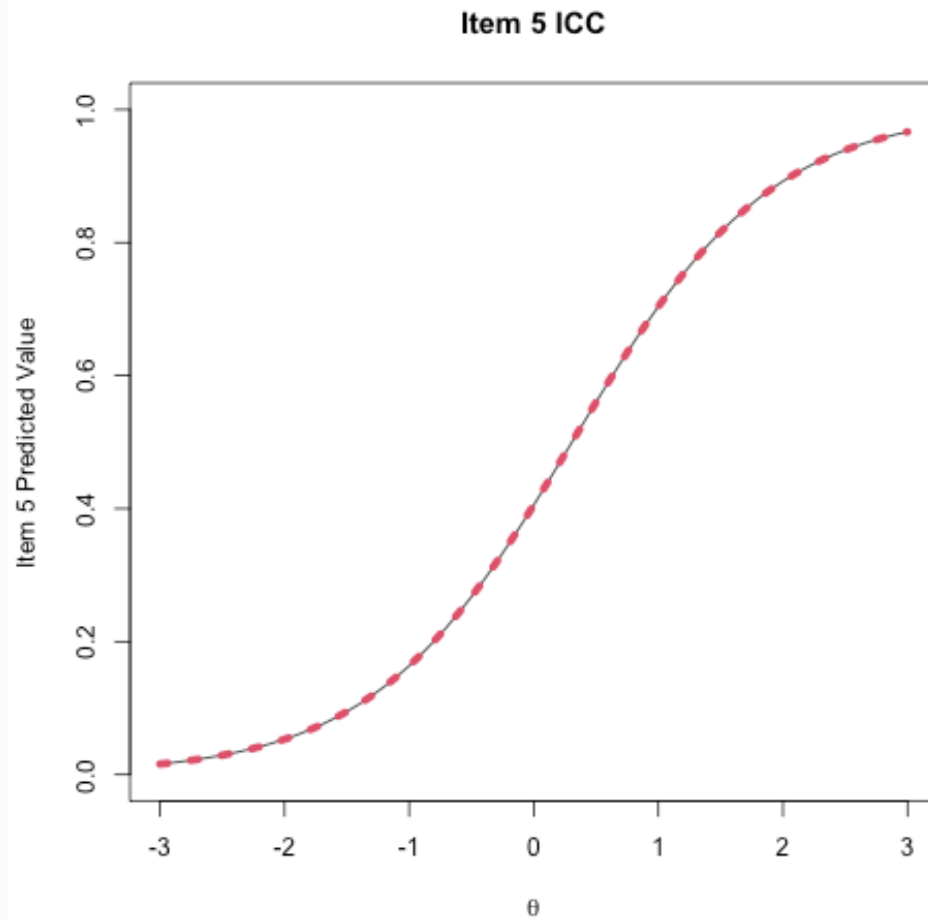
One plot that can help provide information about the item parameters is the item characteristic curve (ICC)

- The ICC is the plot of the expected value of the response conditional on the value of the latent traits, for a range of latent trait values

$$E(Y_{pi} | \theta_p) = \frac{\exp(\mu_i + \lambda_i \theta_p)}{1 + \exp(\mu_i + \lambda_i \theta_p)}$$

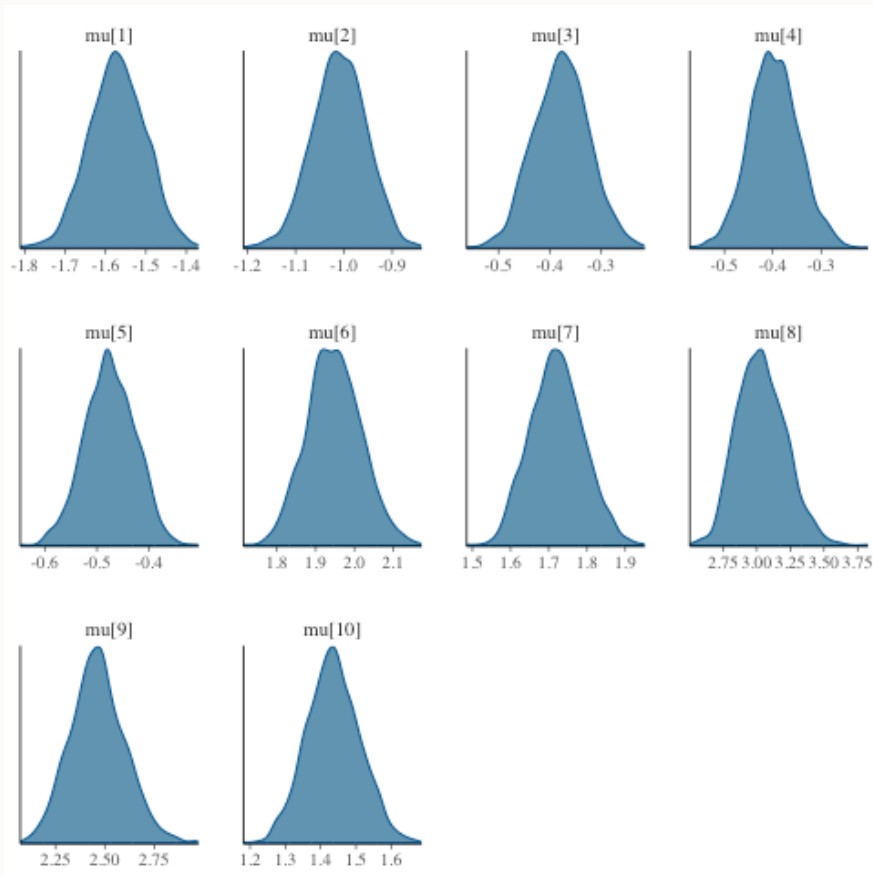
- Because we have sampled values for each parameter, we can plot one ICC for each posterior draw

Posterior ICC Plots



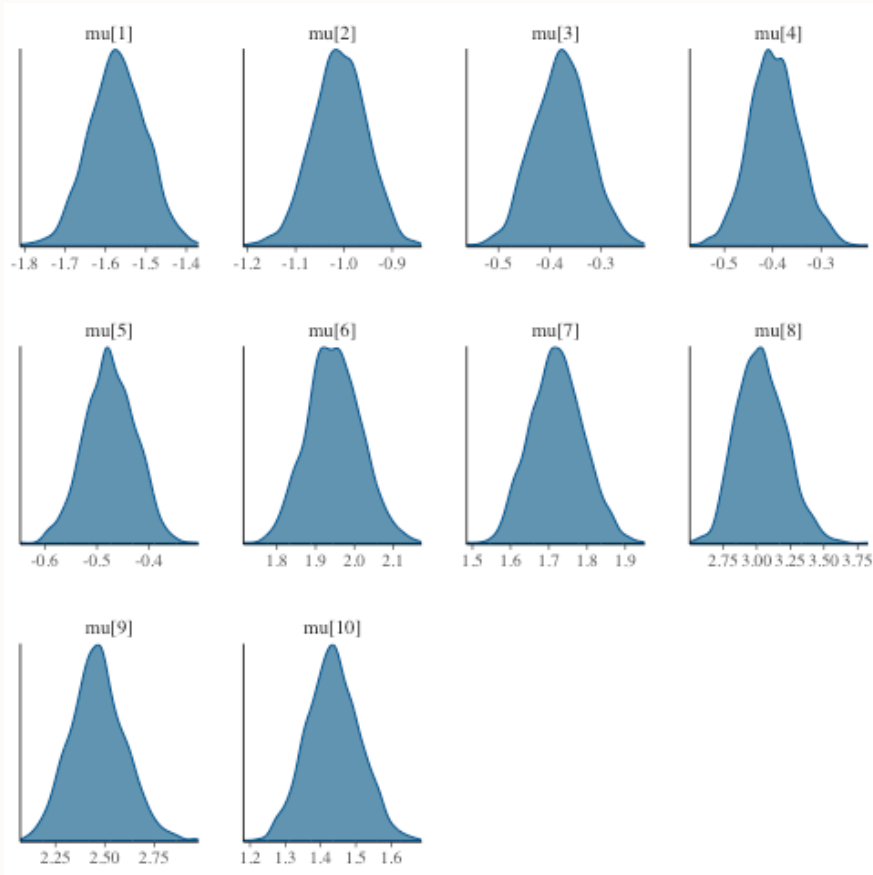
Investigating the Item Parameters

Trace plots for μ_i



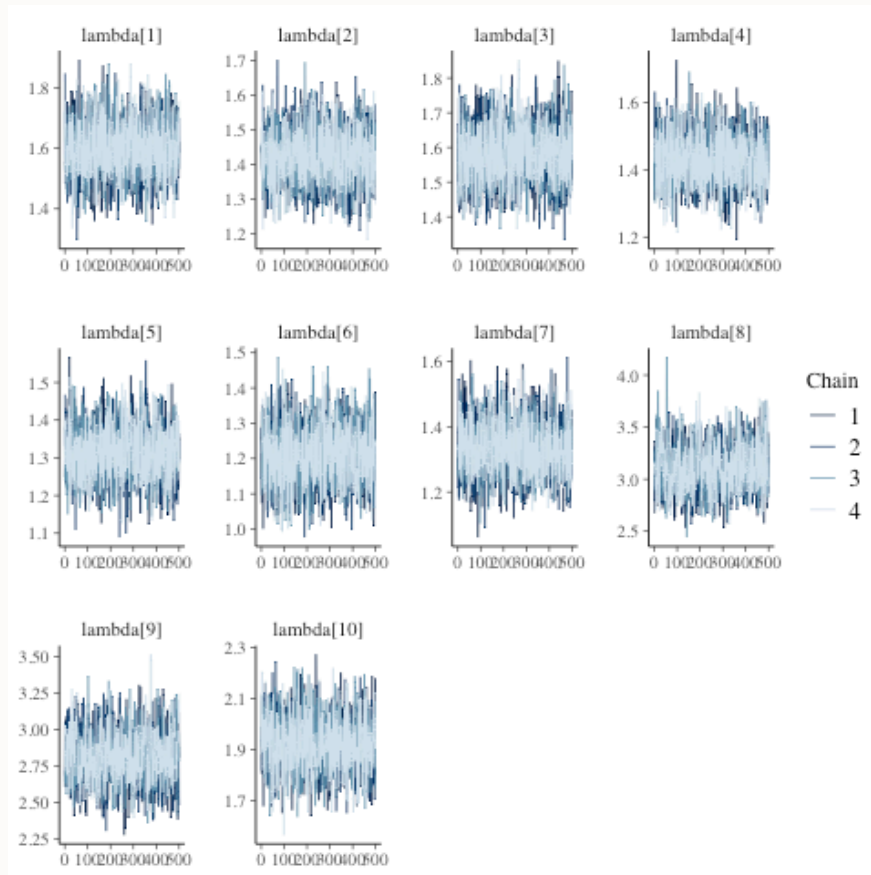
Investigating the Item Parameters

Density plots for μ_i



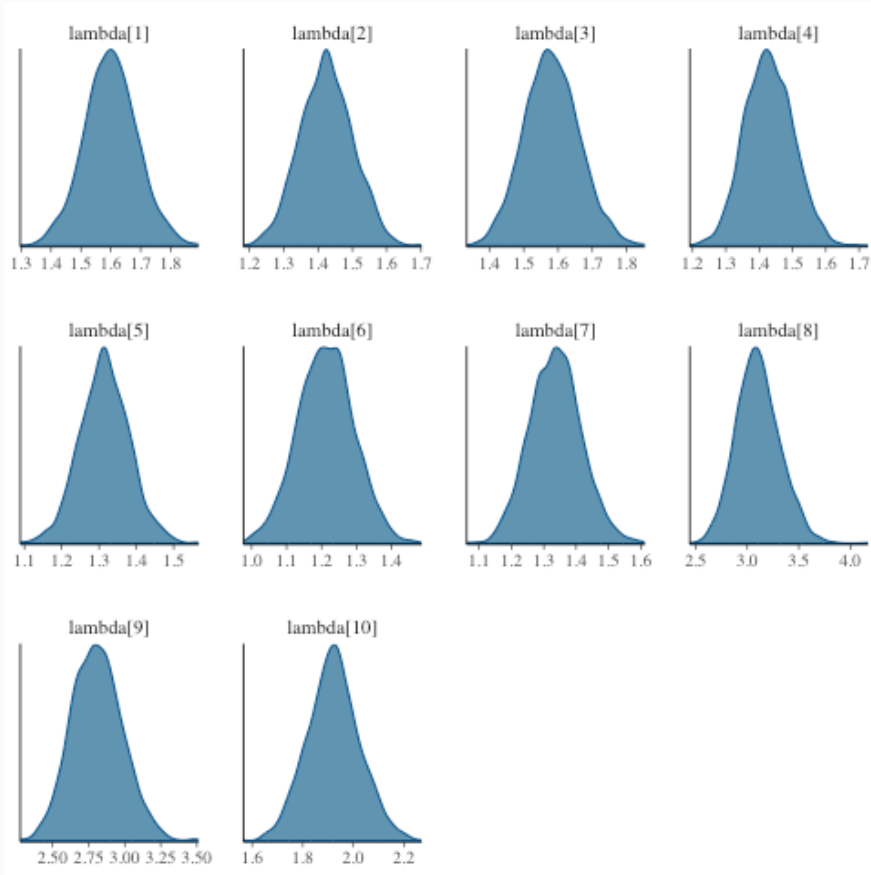
Investigating the Item Parameters

Trace plots for λ_i



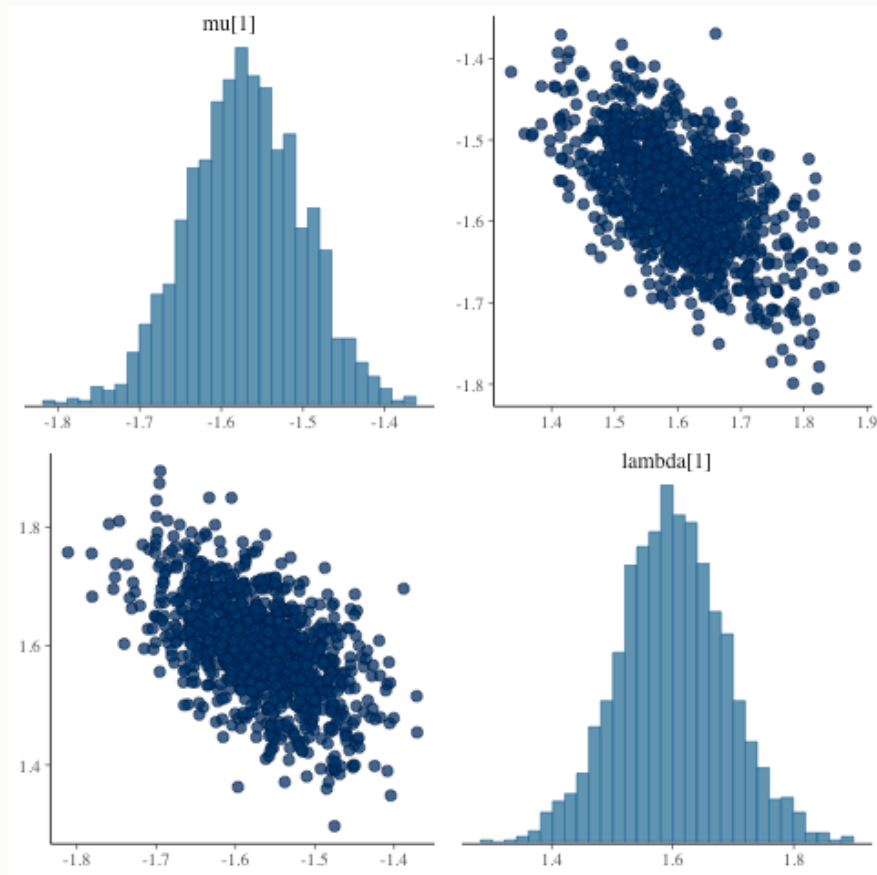
Investigating the Item Parameters

Density plots for λ_i



Investigating the Item Parameters

Bivariate plots for μ_i and λ_i



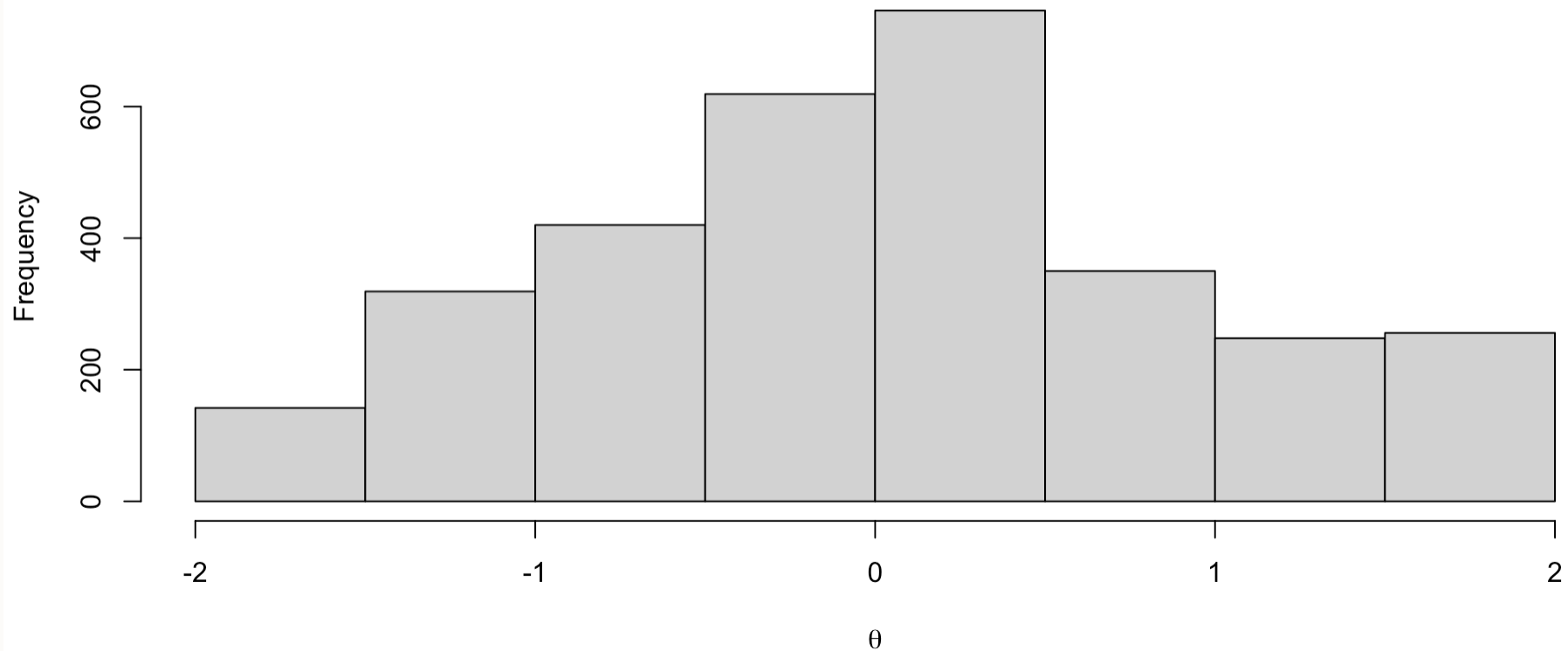
Investigating the Latent Variables

The estimated latent variables are then:

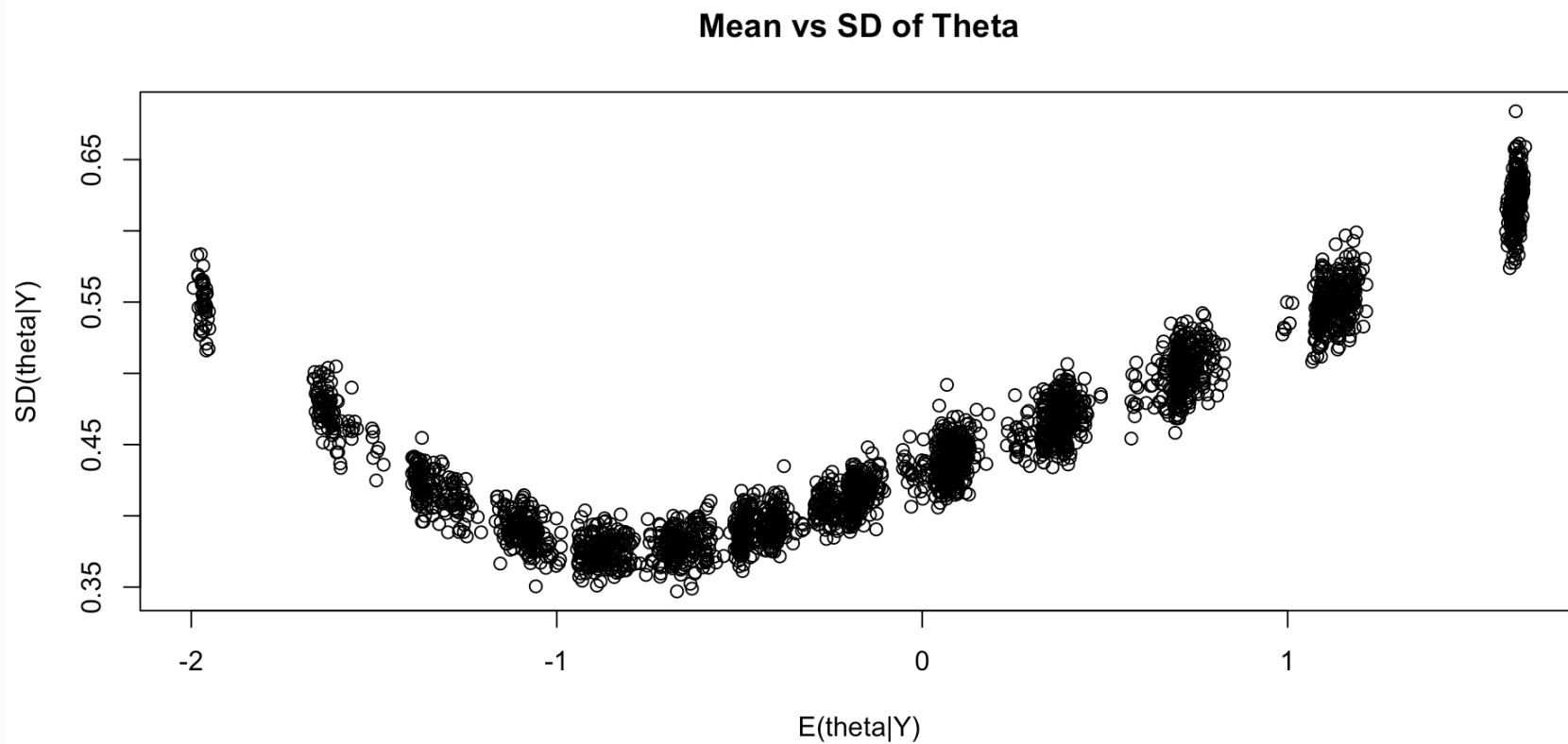
	variable	mean	median	sd	mad	q5
2	theta[1]	-1.6469365380	-1.615670000	0.4812889	0.4792282	-2.489765e+00
3	theta[2]	0.7716262391	0.746348000	0.4928171	0.4947132	-4.661616e-03
4	theta[3]	0.2627107038	0.245656000	0.4449074	0.4608937	-4.729403e-01
5	theta[4]	-0.1693556194	-0.183219000	0.4143666	0.4059782	-8.203617e-01
6	theta[5]	-0.4764400951	-0.489230500	0.3803478	0.3662645	-1.098720e+00
7	theta[6]	-0.6975760584	-0.706521500	0.3868626	0.3745062	-1.342136e+00
8	theta[7]	-1.5571929985	-1.523360000	0.4614712	0.4441573	-2.391607e+00
9	theta[8]	-0.3910862126	-0.390830000	0.3895415	0.3990789	-1.006606e+00
10	theta[9]	0.7024053688	0.688521500	0.4842760	0.4895397	-3.859440e-02
11	theta[10]	-0.6619470321	-0.661569000	0.3855667	0.3861143	-1.282558e+00
12	theta[11]	-0.1901852288	-0.207759000	0.4211212	0.4331752	-8.606453e-01
13	theta[12]	-1.0310495370	-1.020690000	0.3756010	0.3718887	-1.663954e+00
14	theta[13]	-1.6175001030	-1.585010000	0.4806094	0.4643800	-2.447601e+00
15	theta[14]	0.7321750624	0.715731000	0.5207530	0.5441009	-5.309056e-02
16	theta[15]	-0.8100219981	-0.816600000	0.3830759	0.3810238	-1.431341e+00
17	theta[16]	-0.1647927264	-0.172181000	0.4083666	0.4215262	-8.145184e-01
18	theta[17]	0.0483657493	0.038876750	0.4268612	0.4441881	-6.146646e-01
19	theta[18]	-0.1403626413	-0.151256500	0.4229847	0.4121420	-8.339441e-01
20	theta[19]	0.3954864970	0.383822500	0.4810726	0.4642995	-3.596969e-01
21	theta[20]	1.6151871395	1.568615000	0.6202673	0.6145599	6.715304e-01
22	theta[21]	-0.6932722959	-0.699686000	0.3721689	0.3690169	-1.285315e+00
23	theta[22]	0.0677191854	0.056845900	0.4238480	0.4310104	-6.131203e-01
24	theta[23]	0.7700454544	0.750525500	0.4901850	0.4690457	4.446797e-03
25	theta[24]	-0.4917464543	-0.501833000	0.3612975	0.3474399	-1.074091e+00
26	theta[25]	0.0991475174	0.073921800	0.4365510	0.4506882	-5.751500e-01
27	theta[26]	1.6347601445	1.609530000	0.6314360	0.6017503	6.573469e-01
28	theta[27]	1.1819010034	1.155135000	0.5488344	0.5504746	3.296031e-01
29	theta[28]	0.0985017078	0.095088950	0.4362795	0.4328302	-5.803616e-01
30	theta[29]	0.0799025742	0.066492200	0.4414475	0.4513645	-6.222075e-01
31	theta[30]	0.0653543465	0.052225800	0.4261223	0.4482352	-6.154090e-01
32	theta[31]	1.0898006278	1.065960000	0.5578166	0.5502662	2.469389e-01
33	theta[32]	0.7038166019	0.697235500	0.4814161	0.4870430	-8.877802e-02
34	theta[33]	0.0745863020	0.056424250	0.4549470	0.4488090	-6.537972e-01
35	theta[34]	0.7003674712	0.682907500	0.4973362	0.5095489	-8.727124e-02
36	theta[35]	-1.6266668845	-1.596165000	0.4839911	0.4427488	-2.497417e+00
37	theta[36]	-0.4352690085	-0.449557500	0.3913395	0.3927237	-1.055520e+00
38	theta[37]	0.3647141797	0.338409500	0.4664230	0.4663145	-3.703658e-01
39	theta[38]	-0.4709355853	-0.481340000	0.3708361	0.3720370	-1.059918e+00
40	theta[39]	-1.3367518717	-1.308265000	0.4326597	0.4234231	-2.069435e+00
41	theta[40]	-1.9697455105	-1.908580000	0.5629290	0.5400741	-2.984310e+00
42	theta[41]	1.1313906156	1.095450000	0.5905510	0.5758493	2.437457e-01
43	theta[42]	-0.1177583680	-0.128825500	0.4197405	0.4068521	-7.878281e-01
44	theta[43]	-0.4625494286	-0.470941000	0.4012854	0.3892996	-1.102953e+00

EAP Estimates of Latent Variables

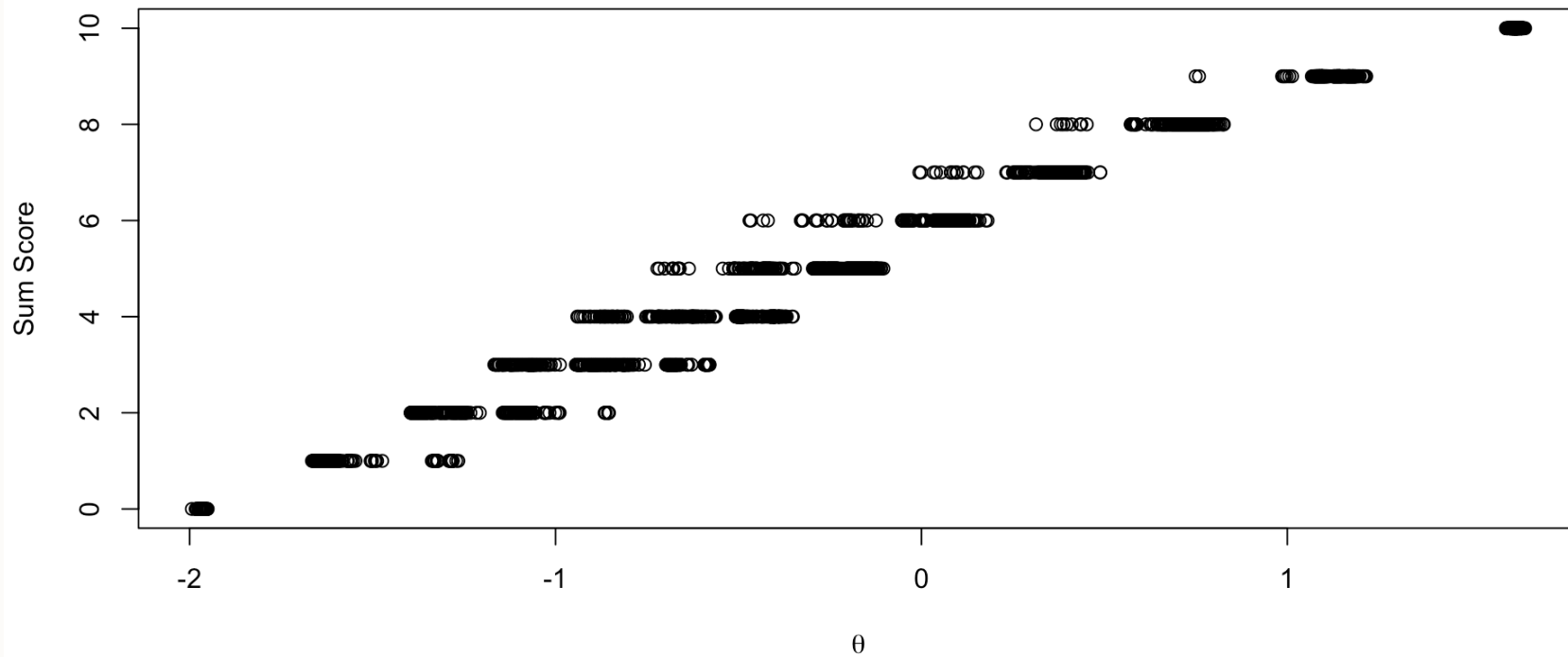
EAP Estimates of Theta



Comparing EAP Estimates with Posterior SDs



Comparing EAP Estimates with Sum Scores



Additional IRT Information

There is a lot more to IRT models, but most is tangential to our workshop

- See Bayesian Psychometric Models my course notes at
 - 2024 (with Stan): <https://jonathantemplin.com/bayesian-psychometric-modeling-fall-2024/>
 - 2019 (with JAGS): <https://jonathantemplin.com/bayesian-psychometric-modeling-spring-2019/>

Posterior Predictive Model Checking for Absolute Fit in Bayesian Psychometric Models

Bayesian Psychometric Model Fit Methods

Objectives:

1. Show how to use PPMC to evaluate absolute model fit in Bayesian psychometric models
2. Show how to use LOO and WAIC for relative model fit in Bayesian psychometric models

Psychometric Model PPMC

Psychometric models can use posterior predictive model checking (PPMC) to assess how well they fit the data in an absolute sense

- At each iteration of the chain, each item is simulated using model parameter values from that iteration
- Summary statistics are used to evaluate model fit

Psychometric Model PPMC

Types of statistics:

- Univariate measures (each item, separately):
 - Item mean
 - Item χ^2 (comparing observed data with each simulated data set)
- Not very useful unless:
 - Parameters have “obscenely” informative priors
 - There are cross-item constraints on some parameters (such as all loadings are equal)
- Bivariate measures (each pair of items)
 - For binary data: Tetrachoric correlations
 - For polytomous data: Polychoric correlations (but difficult to estimate with small samples), pearson correlations
 - For other types of data: Pearson correlations

Problems with PPMC

Problems with PPMC include

- No uniform standard for which statistics to use
 - Tetrachoric correlations? Pearson correlations?
- No uniform standard by which data should fit, absolutely
 - Jihong Zhang has some work on this topic, though:
 - Paper in *Structural Equation Modeling*
 - Dissertation on PPMC with M2 statistics (working on publishing)
- No way to determine if a model is overparameterized (too complicated)
 - Fit only improves to a limit

Implementing PPMC in Stan (one θ)

```
1 generated quantities{
2
3   // for PPMC:
4   array[nItems, nObs] int<lower=0> simY;
5
6   for (item in 1:nItems){
7     for (obs in 1:nObs){
8       // generate data based on distribution and model
9       simY[item, obs] = bernoulli_logit_rng(mu[item] + lambda[item]*theta[obs]);
10    }
11  }
12 }
13 }
```

Notes:

- Generated quantities block is where to implement PPMC
- Each type of distribution also has a random number generator
 - Here, `bernoulli_logit_rng` goes with `bernoulli_logit`
- Each may have some issue in types of inputs (had to go person-by-person in this block)
- Rather than have Stan calculate statistics, I will do so in R

PPMC Processing

Stan generated a lot of data—but now we must take it from the format of Stan and process it:

- For this, we refer to the file `lecture05_unidimensionalBayesianIRT.R`
- Each IRT (or ML IRT) model has a section for PPMC
- “Helper” function at top of syntax to calculate tetrachoric correlation between items

Relative Model Fit in Bayesian Psychometric Models

Relative Model Fit in Bayesian Psychometric Models

As with other Bayesian models, we can use WAIC and LOO to compare the model fit of two Bayesian models

- Of note: There is some debate as to whether or not we should marginalize across the latent variables
 - We won't do that here as that would involve a numeric integral
- What is needed: The conditional log likelihood for each observation at each step of the chain
 - Here, we have to sum the log likelihood across all items
 - There are built-in functions in Stan to do this
- Each IRT (or ML IRT) model has a section for LOO/WAIC

Implementing ELPD in Stan (one θ)

```
1 generated quantities{
2
3   // for LOO/WAIC:
4   vector[nObs] personLike = rep_vector(0.0, nObs);
5
6   for (item in 1:nItems){
7     for (obs in 1:nObs){
8       // calculate conditional data likelihood for LOO/WAIC
9       personLike[obs] =
10         personLike[obs] +
11         bernoulli_logit_lpmf(Y[item, obs] | mu[item] + lambda[item]*theta);
12     }
13   }
14 }
```

Notes:

- `bernoulli_logit_lpmf` needs the observed data to work (first argument)
- `vector[nObs] personLike = rep_vector(0.0, nObs);` is needed to set the values to zero at each iteration prior to summing

Section Summary

Model fit is complicated for psychometric models

- Bayesian model fit methods are even more complicated than non-Bayesian methods
- Open area for research!